

Trivision Kinetic Sculpture Manual

last revision Nov. 29, 2025

Status: Design & Specification Phase

I. Project Overview

This project details the design and fabrication of a parametric kinetic sculpture based on a "Trivision Billboard" (Rotating Agamograph) mechanic. It is a mechanical display for three distinct images, interlaced and mapped onto 12 rotating triangular prisms.

Primary prism starting size will be **4.25" wide x 36" tall** (52" wide total for 12 prisms, and 62" wide artwork)

5.667" wide faces with **48" tall** prisms for a giant museum-sized piece

Any 1:8.5 aspect ratio size strip size will work and can be designed by scaling the design parametrically.

Core Aesthetic: The motion profile is a "cascading wave" but can be programmed into other patterns. The prisms trigger sequentially (Prism 1, then Prism 2, etc.), creating a waterfall effect.

II. High-Level Strategy

1. Parametric Design Approach A custom software workflow will generate fabrication data based on variable inputs (Image size, Prism count). This ensures the design is scalable and repeatable.

2. Direct Drive Topology The system will use 12 independent stepper motors with programmable software acceleration curves. Hall Effect sensors will allow homing to zero position to correct any drift.

3. Silent Operation The installation is designed for gallery environments. Hardware selection prioritizes silent operation through "SilentStepStick" drivers, fanless power supplies, and high-resolution motors to eliminate vibration whine.

4.. Power Safety & Management

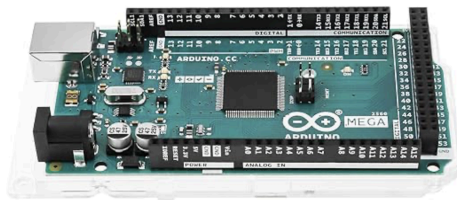
- Running the motors at **24V** ensures high torque and speed.

- The Arduino cannot handle 24V directly and will be wired to 24V through an intermediate **DC-DC Buck Power Converter** 4-40V to 1.25-37V to step voltage down to 9V for the logic board. 9V allows headroom for brownouts when the motors kick in.
- *Headroom:* The system is calculated to draw ~80W under normal load. The 150W PSU provides ~50% safety headroom, ensuring cool, fanless operation.

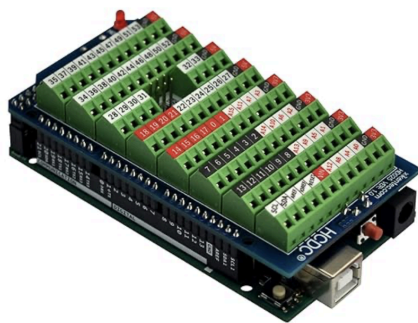
III. Hardware Bill of Materials (BOM)

1. Electronics & Control

- **Microcontroller:** Arduino Mega 2560 R3 (Selected for 54 digital I/O pins to manage 12 sensors, probably 3 wires each, STEP, DIR, and EN)



- A **Screw Terminal Block Breakout Shield Module** on top will mean no soldering.

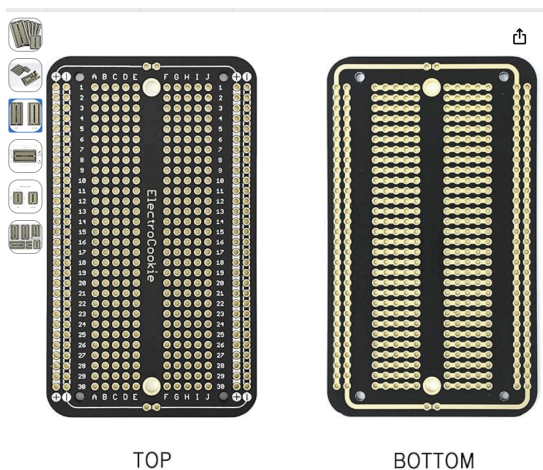


- **Motor Drivers:** Adafruit TMC2209 Stepper Motor Driver Breakout Board, with a 10-pin terminal block soldered on so wires can be screw-attached.

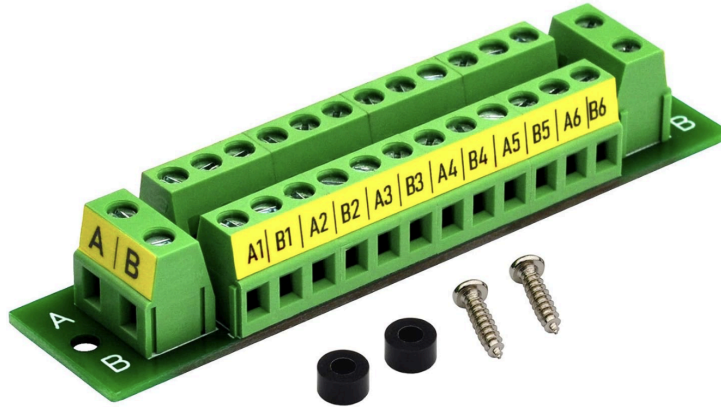
- **Voltage Regulation (Logic):** LM2596 DC-DC Buck Converter with LED Voltmeter Display (Steps 24V down to 9V for Arduino safety).



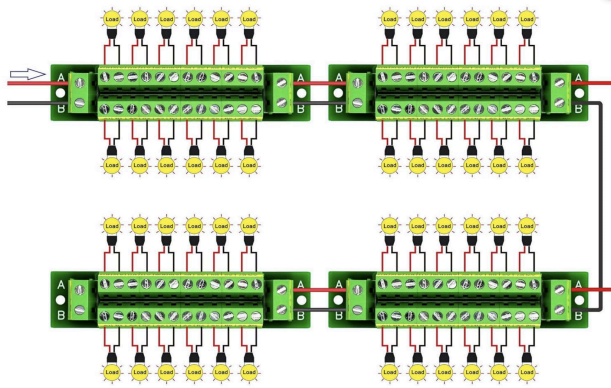
- **5V and GND from all 12 motors** to be soldered to a common “bus” across the + and - rails on an Electrocookie PCB Prototyping Board. Gemini suggests I *might* need to install a **Noise Filter**: a **100μF to 470μF Electrolytic Capacitor** on the power rails at the "head" of the board (e.g., Row 1). **Polarity:** Ensure the capacitor's Negative stripe faces the GND rail. This would act as a localized reservoir, smoothing out the voltage dips caused when 12 motors accelerate simultaneously.



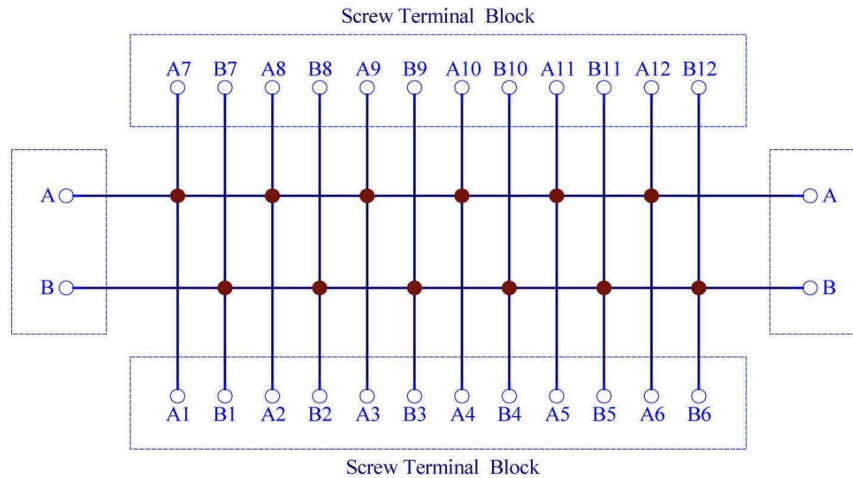
16 Amp 2x12 Position Terminal Block Distribution Module
for AC DC



Typical Connection Use Method



Schematic



2. Motion System

- **Motors:** StepperOnline 17HM19-2004S Nema 17 Stepper Motors.
 - *Spec:* 0.9° Step Angle (High Resolution), 46Ncm Torque, 2.0A Rated Current.
- **Homing Sensors:** A3144 Hall Effect Sensors (TO-92 Package). *To be rigged in custom 3D mounts that clip onto the motors.*

3. Power System

- **Primary Power Supply:** Mean Well LRS-150-24.
 - *Spec:* 24V DC, 6.5A, 150W.
 - *Feature:* Fanless/Convection Cooled (Silent).

4. Wiring Plan

AC power goes:

Black > LRS-150-24 Power Supply "L"
White > LRS-150-24 Power Supply "N"
Green > LRS-150-24 Power Supply "Ground" (icon)

Then:

-V from LRS-150-24 > Motor Driver -
+V from LRS-150-24 > Motor Driver +

Then:

-V from LRS-150-24 > Buck Converter VIN-
+V from LRS-150-24 > Buck Converter VIN+

Then:

Buck Converter VOUT- > Arduino GND
Buck Converter VOUT+ > Arduino VIN

TRIVISION PIN ASSIGNMENTS

Motor STEP pins (pins 35–46)

Motor 1: STEP pin 35 Motor 2: STEP pin 36 Motor 3: STEP pin 37 Motor 4: STEP pin 38 Motor 5: STEP pin 39 Motor 6: STEP pin 40 Motor 7: STEP pin 41 Motor 8: STEP pin 42 Motor 9: STEP pin 43 Motor 10: STEP pin 44 Motor 11: STEP pin 45 Motor 12: STEP pin 46

Motor DIR pins (pins 18–29)

Motor 1: DIR pin 18 Motor 2: DIR pin 19 Motor 3: DIR pin 20 Motor 4: DIR pin 21 Motor 5: DIR pin 22 Motor 6: DIR pin 23 Motor 7: DIR pin 24 Motor 8: DIR pin 25 Motor 9: DIR pin 26 Motor 10: DIR pin 27 Motor 11: DIR pin 28 Motor 12: DIR pin 29

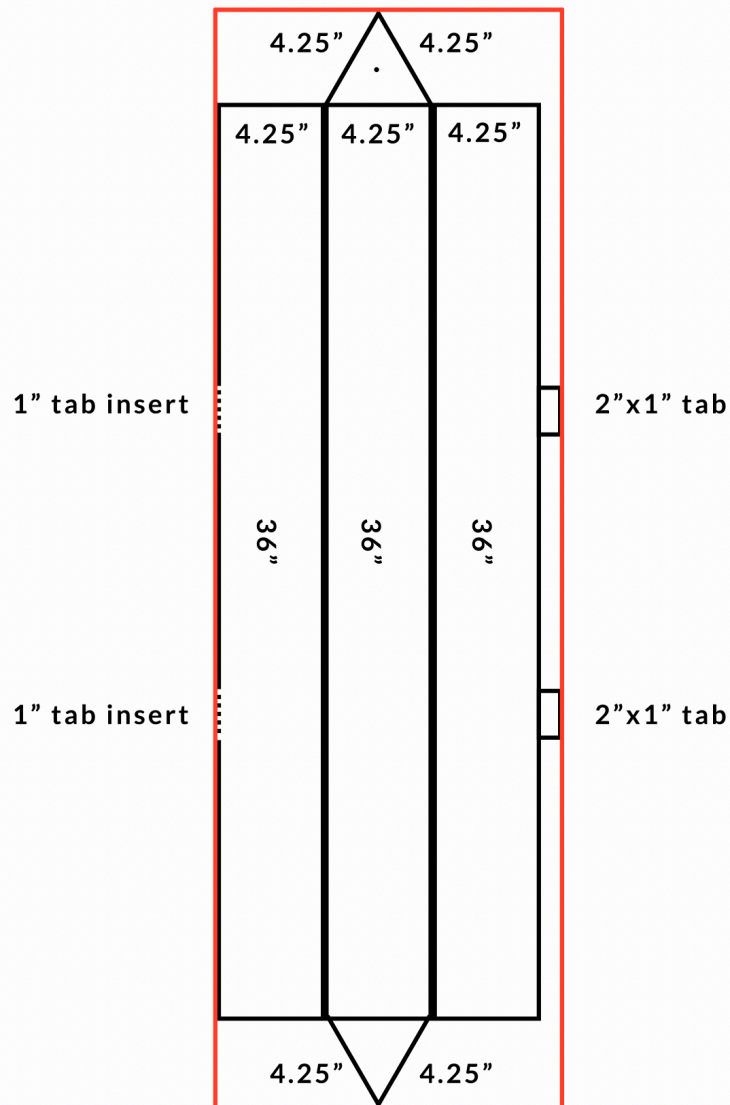
Hall Effect sensor pins (pins 2–13)

Motor 1: Hall pin 2 Motor 2: Hall pin 3 Motor 3: Hall pin 4 Motor 4: Hall pin 5 Motor 5: Hall pin 6 Motor 6: Hall pin 7 Motor 7: Hall pin 8 Motor 8: Hall pin 9 Motor 9: Hall pin 10 Motor 10: Hall pin 11 Motor 11: Hall pin 12 Motor 12: Hall pin 13

5. Fabrication Materials

- **Prism Substrate:** 1/8" (0.118") Dibond (Aluminum Composite Material, ACM) - Matte White.
 - *Source:* US Plastic Corp, Item #155783 (48" x 96" sheet).
 - **Prism Structure:** 3-face hollow assembly with 3D printed top/bottom end-caps.
 - Shafts will weigh about 4 lbs each and to avoid axial load on the steppers which are not rated for axial loads, I will need to transfer load onto stepper with axial thrust bearings (in Amazon cart)
 - Next time you can get Stephen at Pegleg to cut it on his CNC.
 - Ordering 8 black sheets, from
<https://www.usplastic.com/catalog/item.aspx?itemid=155783>

CNC plunge cut (after shallow fold cut)
to something like 13.75x43"



Want to put the prisms on a rectangle

Will need a 120° V-bit to router-cut the folds (outer sandwich layer of dibond)

12/11/25 ordered: Amana Tool RC-1146 CNC Insert V-Groove 120 Deg x 1-23/64 D x 25/64
CH x 1/4 SHK Router Bit

Z-Zero Strategy: Do not set your Z-zero to the top of the material. Set it to the **spoilboard (machine bed)**.

- Your cut depth should be set to leave exactly **0.025" - 0.030" (0.6mm - 0.8mm)** of material remaining.



- **Steel Shafts:** 52" shafts (up to 56" is OK)

The Motor shaft is 5mm. The linear motion rod is 6mm. I will be using a 5mm to 6mm coupler: 5mm to 6mm Clamping Rigid Coupling, Set Screw Coupler



- **Image Media:** Hahnemuhle Baryta Luster paper (Adhered to ACM faces).

- **Linear Motion Rods:**

- **Framing material:**

6/4 × 7.5” white oak, surfaced (S4S):

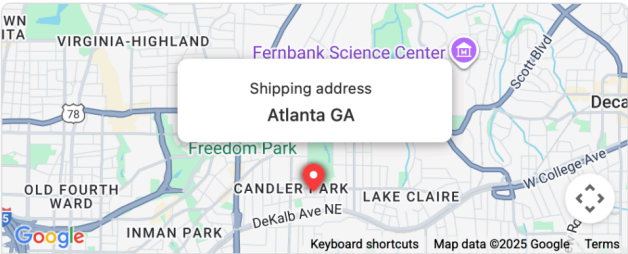
You really need a minimum of 7.5” or you can’t get the guts into it:

6/4 means 1.5” rough, surfaced down to 1.25” smooth

Black cut 3mm ACM ordered 12/12/25



Confirmation #0MA5UKJYF
Thank you, Joel!



Your order is confirmed

You'll receive a confirmation email soon

Order details

Contact information

joel@silvermanphoto.com

Shipping address

Joel Silverman
Joel Silverman Studio
1501 Iverson Street
Atlanta GA 30307
United States
4049648236

Shipping method

Shipping Zone 1 - 4

Payment method

PayPal · \$726.07 USD

Billing address

Joel Silverman
Joel Silverman Studio
1501 Iverson Street
Atlanta GA 30307
United States
4049648236

sheet_1_custom_size_4: 48x16
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24 Custom Cut Fee \$110.40
productid: 43494921502965
tokenID: 1765517882002

Subtotal · 28 items \$716.08
Shipping ⓘ \$9.99

Total USD **\$726.07**

Motor #1

- VDD (pin 1) → Electrocookie 5V bus **RED**
 - GND (pin 2) → OONO B bus (GND) **YELLOW**
 - DIR (pin 3) → HCDC terminal 18 (column 5) **GREEN**
 - STEP (pin 4) → HCDC terminal 35 (column 8) **WHITE**
 - Hall brown wire (VCC) → Electrocookie 5V bus **BROWN**
 - Hall blue wire (GND) → OONO B GROUND bus or your Electrocookie GND rail **BLUE**
 - Hall black wire (signal) → HCDC terminal 2 (column 3) **BLACK**
-

Motor #2

- VDD (pin 1) → Electrocookie 5V bus
 - GND (pin 2) → OONO B bus (GND)
 - DIR (pin 3) → HCDC terminal 19 (column 5)
 - STEP (pin 4) → HCDC terminal 36 (column 7)
 - Hall brown wire (VCC) → Electrocookie 5V bus
 - Hall blue wire (GND) → OONO B GROUND bus or your Electrocookie GND rail
 - Hall black wire (signal) → HCDC terminal 3 (column 3)
-

Motor #3

- VDD (pin 1) → Electrocookie 5V bus
 - GND (pin 2) → OONO B bus (GND)
 - DIR (pin 3) → HCDC terminal 20 (column 5)
 - STEP (pin 4) → HCDC terminal 37 (column 8)
 - Hall brown wire (VCC) → Electrocookie 5V bus
 - Hall blue wire (GND) → OONO B GROUND bus or your Electrocookie GND rail
 - Hall black wire (signal) → HCDC terminal 4 (column 3)
-

Motor #4

- VDD (pin 1) → Electrocookie 5V bus
- GND (pin 2) → OONO B bus (GND)
- DIR (pin 3) → HCDC terminal 21 (column 5)
- STEP (pin 4) → HCDC terminal 38 (column 7)
- Hall brown wire (VCC) → Electrocookie 5V bus
- Hall blue wire (GND) → OONO B GROUND bus or your Electrocookie GND rail

- Hall black wire (signal) → HCDC terminal 5 (column 3)
-

Motor #5

- VDD (pin 1) → Electrocookie 5V bus
 - GND (pin 2) → OONO B bus (GND)
 - DIR (pin 3) → HCDC terminal 22 (column 5)
 - STEP (pin 4) → HCDC terminal 39 (column 8)
 - Hall brown wire (VCC) → Electrocookie 5V bus
 - Hall blue wire (GND) → OONO B GROUND bus or your Electrocookie GND rail
 - Hall black wire (signal) → HCDC terminal 6 (column 3)
-

Motor #6

- VDD (pin 1) → Electrocookie 5V bus
 - GND (pin 2) → OONO B bus (GND)
 - DIR (pin 3) → HCDC terminal 23 (column 5)
 - STEP (pin 4) → HCDC terminal 40 (column 7)
 - Hall brown wire (VCC) → Electrocookie 5V bus
 - Hall blue wire (GND) → OONO B GROUND bus or your Electrocookie GND rail
 - Hall black wire (signal) → HCDC terminal 7 (column 3)
-

Motor #7 ← Hall sensors switch to column 2 starting here

- VDD (pin 1) → Electrocookie 5V bus
 - GND (pin 2) → OONO B bus (GND)
 - DIR (pin 3) → HCDC terminal 24 (column 5)
 - STEP (pin 4) → HCDC terminal 41 (column 8)
 - Hall brown wire (VCC) → Electrocookie 5V bus
 - Hall blue wire (GND) → OONO B GROUND bus or your Electrocookie GND rail
 - Hall black wire (signal) → HCDC terminal 8 (column 2)
-

Motor #8

- VDD (pin 1) → Electrocookie 5V bus
- GND (pin 2) → OONO B bus (GND)

- DIR (pin 3) → HCDC terminal 25 (column 5)
 - STEP (pin 4) → HCDC terminal 42 (column 7)
 - Hall brown wire (VCC) → Electrocookie 5V bus
 - Hall blue wire (GND) → OONO B GROUND bus or your Electrocookie GND rail
 - Hall black wire (signal) → HCDC terminal 9 (column 2)
-

Motor #9

- VDD (pin 1) → Electrocookie 5V bus
 - GND (pin 2) → OONO B bus (GND)
 - DIR (pin 3) → HCDC terminal 26 (column 5)
 - STEP (pin 4) → HCDC terminal 43 (column 8)
 - Hall brown wire (VCC) → Electrocookie 5V bus
 - Hall blue wire (GND) → OONO B GROUND bus or your Electrocookie GND rail
 - Hall black wire (signal) → HCDC terminal 10 (column 2)
-

Motor #10

- VDD (pin 1) → Electrocookie 5V bus
 - GND (pin 2) → OONO B bus (GND)
 - DIR (pin 3) → HCDC terminal 27 (column 5)
 - STEP (pin 4) → HCDC terminal 44 (column 7)
 - Hall brown wire (VCC) → Electrocookie 5V bus
 - Hall blue wire (GND) → OONO B GROUND bus or your Electrocookie GND rail
 - Hall black wire (signal) → HCDC terminal 11 (column 2)
-

Motor #11 ← DIR switches to column 6 starting here

- VDD (pin 1) → Electrocookie 5V bus
 - GND (pin 2) → OONO B bus (GND)
 - DIR (pin 3) → HCDC terminal 28 (column 6)
 - STEP (pin 4) → HCDC terminal 45 (column 8)
 - Hall brown wire (VCC) → Electrocookie 5V bus
 - Hall blue wire (GND) → OONO B GROUND bus or your Electrocookie GND rail
 - Hall black wire (signal) → HCDC terminal 12 (column 2)
-

Motor #12

- VDD (pin 1) → Electrocookie 5V bus
 - GND (pin 2) → OONO B bus (GND)
 - DIR (pin 3) → HCDC terminal 29 (column 6)
 - STEP (pin 4) → HCDC terminal 46 (column 7)
 - Hall brown wire (VCC) → Electrocookie 5V bus
 - Hall blue wire (GND) → OONO B GROUND bus or your Electrocookie GND rail
 - Hall black wire (signal) → HCDC terminal 13 (column 2)
-

Pins 5–10 on every driver (MS1, MS2, DIAG, INDEX, UART, EN) left unconnected.